1) DIFF BETWEEN ROWID AND ROWNUMBER

ROWID	ROW NUMBER
Row id is a static value (always same row has same rowed)	Row number is not a static values(row number changes dynamically based on select statement)
Row id is hexadecimal value	Row number is numeric value
Row id is combination of object id, file number, block number, row number	Row number is temporary number based on select statement
Oracle automatically generate row id when user insert new row into table	Row number will be generated based output of select statement
Row id will be stored in database	Row number won't be stored in the database

2) Diff b/w truncate and delete

TRUNCATE	DELETE
Truncate removes all the rows from the table	Delete removes all rows and selected rows from the table
Truncate won't support where clause	Delete supports where clause
Truncate is ddl command	Delete is dml command
After truncate we don't need to give and tcl command as truncate is ddl command and all ddl commands are auto commit	After delete we need to give tcl command like commit or rollback
Truncate is faster than delete as it doesn't have undo segment	Delete is slower as it uses undo segment
Truncate clears the memory allocated to rows(internal memory will be cleared)	Delete won't clear the internal memory and internal memory will be still exists
Truncate resets the high level water mark and reclaims the space used by the table	Delete won't reset the high level water mark and won't reclaims the space used by the table

3) Diff between case and decode

CASE	DECODE
Case is statement and it provides if else type logic	Decode is function and it provides if else type logic
Case can be used along with sql statements and without sql statements also we can use case	Decode can be used in sql statements only
Case supports operators like <,>,<=,>=,like, in, between, not	Decode won't support any operators
By using case we can assign any value to a variable in the plsql block	Decode can't be used inside plsql block
Case allows only one data type in single statement	Decode allows any no of data types in single statement
Case is easy to understand	Decode is difficult to understand

4) Diff between where and having

WHERE	HAVING
Where clause is used to get specific data from the table	Having clause is used to apply conditions on grouped data
Where clause filters row data	Having clause filters grouped data
Where conditions are applied before group by clause	Having conditions are applied after group by clause
Where won't support aggregate functions	Having clause supports aggregate functions
After where clause we will write normal columns only	After having clause we can write aggregate functions only
By using where clause we can join any number of tables	By using having clause we can't join tables

5) Diff between table and view

TABLE	VIEW
Table is combination of rows and columns	View is stored select statement
Table occupy some space in database	View won't occupy any space in database
Table stores some data in their self	View is a logical component(it won't occupy any space and stores only select statement)
Table is the preliminary storage object for storing data and information in RDBMS	View won't contain a set of values in database
Table is collection of related data and it consists of rows and columns	View is virtual table and its content depends on the query
If we apply and dml command on related view, it will impact the view	If we apply any dml command on related table, it will impact the table
If drop the related view, it won't impact the table	If we drop the related table, it will impact the view and view will become invalid
Table can have maximum of 1000 columns	View can have maximum of 999 columns
To see tables in our schema: Select * from user_tabs	To see views in our schema: Select * from user_views

6) Diff between synonym and view

SYNONYM	VIEW
Synonym will hide the original name of the object. It is alias name of the object.	View is stored select statement
Synonym will be created on only one object	View can be created and any no. of objects
Synonym will be created on tables, views, functions, procedures…	Views can be created on tables, views, synonyms
To see synonyms in our schema: Select * from user_synonyms	To see views in our schema: Select * from user_views
If we drop the related table, then synonym will become invalid and if we recreate the table	If we drop the related table view will become invalid and if we recreate the table then also

then synonym will become valid. No need to recreate or recompile the synonym	view will be in invalid state. If want that view then we need to recompile or recreate the view.

7) A) What is group by clause?

Group by clause is used to group row based on specific column.

If want to write normal column along with aggregate function in select statement, then that normal column should come after group by clause in that select statement.

B) What is order by clause?

Order by clause is used to arrange select statement output in ascending order or descending order.

By default order by clause has ascending order. If want to arrange select statement output in descending order then after order by clause we need to write desc keyword.

We can write any no of columns in order by clause but make sure that the column that is used to sort, that column will exist in the column_list of the select statement.

C) What is having clause?

Having clause is used to apply conditions to grouped data.

Having clause filters grouped data.

Having conditions are written after group by clause.

Having clause always follows group by clause.

After having clause we must write aggregate functions only.

8) What is view and what are the advantages of views?

View is stored select statement. View won't store any data physically on it. View won't occupy any space in the database.

Views can be created on tables, views, synonyms.

Any dml operation on the view will impact the original object.

If we drop the related table, view will be invalid. And if we recreate the table even then also view will be in invalid state only. If we want that view then we need to recreate or recompile it.

Advantages of views:

- View will hide the complexity of the query.
- It supports and provides high security while sharing selected rows/information to other users.
- If we want to share data of more tables then we can do it simply by using as single view.
- When the frontend resource uses the same select statement multiple times and he needs some time for it to exist in the frontend. So, instead of that we will create a view on that query.

9) Why sequence is required? What are the advantages and disadvantages of sequence?

Sequence is required to generate unique and order wise values.

Sequence consists of start with, increment by, minimum value, maximum value, cycle, cache.

Advantages of sequence:

- No need to create any table for sequence.

- If the related table is dropped, it won't impact the sequence.
- Sequence gives unique and order wise values.
- Sequence won't belong to any object.

Disadvantages of sequence:

- We cannot rollback sequence (we cannot get the start value again once the sequence is started.
- Sequence doesn't belong to any object.

To get the next value of sequence:
Select seq_name.nextval from dual;
To get the current value of sequence:
Select seq_name.curval from dual;

10) What is synonym and what are the advantages of synonym?

- Synonym is used to share information of an object of one user to other user. And both the users should be connected to same database.
- Synonyms can be created on tables, views, functions, procedures.
- One synonym can be used for only one object.
- Synonym hides the original name of the object.
- We can create synonym on another synonym.

Synonym provides high security while sharing information to other users.

Synonyms are of two types.

They are 1) Private synonyms: select * from user_synonyms;
2) Public synonyms: select * from all_synonyms;

If the related table is dropped then synonym will become invalid. And if the table is recreated then the synonym will be in valid state. No need to recompile or recreate it.

Synonym is nothing but alias name of the original object. Whatever changes are done on synonym that will impact on the original object and whatever changes are done on the original object that will impact on the synonym.

11) Explain about each constraint./ what is constraint and what are its types?

Constraints mean rules or regulations.
By using constraints we can restrict the table data.
When we want to put any rule or restriction on column date then we can use constraints.
We cannot see or modify the code of constraints, as they are encrypted.
Constraints are of 4 types.

a) Unique key constraint
b) Primary key constraint
c) Check key constraint
d) Foreign key constraint

Unique key constraint:

- When we create ukc then automatically uk index will be created

- Unique won’t allow duplicate values
- Table can have any no of unique key.
- If we create unique constraint on more than one column of a table then it is called composite unique key constraint.

Primary key constraint:

- When we create pkc then automatically unique index will be created
- Pk won’t allow null values and duplicate values.
- Table can have only one primary key
- If you create primary key for more than one columns on table at a time then it is called composite primary key

Check constraint:

- It is used to specify conditional restrictions.
- It will accept both null and duplicate values.
- It is user friendly constraint.
- It won’t create unique index automatically.

Foreign key constraint:

- Used to specify the relation between 2 tables.
- If you want to create a fk then reference table should be primary or unique.
- Fk will accept both null values and duplicate values.
- Fk won’t create index.

12) Diff between unique and primary key.

UNIQUE	PRIMARY
Unique key allows null values	Primary key won’t allow null values (unique + not null = primary)
Table can have any no of unique keys	Table can have only one primary key
Unique index will be created automatically	unique index will be created automatically
Unique won’t allow duplicate values	Primary key won’t allow both duplicate and null values
To make unique key as primary key then we need to change the column having unique key to not allow null values	Primary key always wont allows null values.

13) Diff between primary and foreign key.

PRIMARY	FOREIGN/REFERENCE
Primary key won’t allows duplicate and null values	Foreign key allows null and duplicate values
Table can have only one primary key	Table can have any number of foreign key
Index will be created automatically	Index won’t be created
Primary key can be created on any column	Foreign key can be created on any column but the referenced column must have primary or unique key
Primary key won’t support on delete cascade or on delete set null	Foreign key supports on delete cascade and on delete set null

14) Diff between max and greatest.

MAX	GREATEST
Max is aggregate function	Greatest is arithmetic function
Max picks greatest value from the given column and return single row	Greatest picks the highest value from the given columns and return single column
Max is used to go down the column	Greatest is used across the rows
Max gives value in the single row	Greatest gives multiple rows in a single column from the given multiple columns of data
Max is row level function	Greatest is column level function

15) Diff between char and varchar2.

CHAR	VARCHAR
Char is fixed length data type	Varchar is variable length data type
Char allocates memory irrespective of length of string means if char(10) is there and if we have given a string of 4chars, but char will allocate 10 spaces even for that 4 chars	Varchar allocates memory wrt to length of string means if varchar(10) is there and we have given string of 4chars, varchar allocates only 4 spaces for the string remaining memory it won't waste
Char wastes memory means	Varhcar don't waste memory
Char max size 2000	Varchar max size 4000 >>>11g Varchar max size 32767>>>12c

16) Diff between sql and plsql.

SQL	PL/SQL
Sql is Structured Query Language	Pl/sql is Procedural language extension on sql
Sql directly interacts with database server	Plsql won't interacts directly with database server
Sql won't support for loop, if conditions	Plsql support for loop, if conditions, while loop
Sql won't support variables	Plsql support variables and data types
Sql is data oriented language	Plsql is application oriented language
Sql is mainly used to manipulate data	Plsql is mainly used to create an application
At a time in sql we can execute only one statement	At a time in plsql we can execute multiple statements

17) Diff between named block and unnamed block.

NAMED BLOCK	UN-NAMED BLOCK
Plsql blocks which are having header section are known as named blocks(I.e we can create by using create or replace)	Blocks which are not having header section are called as anonymous blocks/un_named blocks(i.e we can create by using declare, begin, end)
Named blocks are stored in database	Un-named blocks are not stored in database
Named blocks are reusable	Un-named blocks are not reusable
Named blocks return a value(function must return a value)	Un-named blocks won't return a value
We can use parameters inside named blocks	We cannot use parameters inside un-named blocks

We can give permissions to other users on named blocks	We cannot give permissions to other users on un-named blocks
Once we compile named blocks, they will be stored permanently as p_code in the shared pool of sga(system global area)	We need to compile every time when executed
Named blocks can be called anywhere in the schema	Un-named blocks won't call in any other block

18) Diff between procedure and function.

FUNCTION	PROCEDURE
Function can be called inside select statement	Procedure cannot be called inside the select statement
Function must return a value	Procedure may or may not return a value
We can use return statement inside a function	We can use return statement inside a procedure. But that doesn't act like the return statement inside the function
In functions we can use dml operations. But the functions with dml operations cannot be called inside the select statement	In procedures we can use dml operations
If we want call the functions with dml operations inside select statement then we can use pragma autonomous transaction + tcl command	Irrespective of dml operations, we cannot call procedure inside select statement
Out parameter is not required inside the function, as function must return a value	If we want any output in procedure then we can use out parameter
If we want, we can use out parameter in function but that function cannot be called inside the select statement	Irrespective of out parameter , we cannot call procedure inside select statement

19) Diff between constraint and trigger.

TRIGGER	CONSTRAINT
Trigger is user defined	Constraint is pre defined
Trigger are 3 types DDL DML DATABASE	Constraints are 4 types UNIQUE KEY CONSTRAINT PRIMARY KEY CONSTRAINT CHECK CONSTRAINT FOREIGN KEY CONSTRAINT
Triggers can be written on tables, schema, database	Constraints can be created only on columns
Trigger can be seen and modify the code	Constraints code cannot be seen and cannot be modified as they are encrypted
Whatever code we want, that can be written in triggers, user friendly	Constraints always follows same rules and regulations
To see triggers in our schema: Select * from user_triggers	To see constraints in our schema: Select * from user_constraints
Whatever we cannot handle through constraints that can be handled through triggers.	By using constraints we cannot maintain audit_table

i.e audit table can be maintained by using triggers.	
Triggers won't support on delete cascade and on delete set null	Constraint support on delete cascade and on delete set null
Triggers have concept of 'instead of trigger' for complex views	Constraints won't work on views

20) Rank and dense_rank diff

DENSE RANK	RANK
If 5members got same marks then desne rank gives 1st rank those 5members and for the 6th member it will give 2nd rank	If 5members got same marks then rank gives 1st rank those 5members and for the 6th member it will give 6th rank
Last value in dense rank column will be equal to no of records in the table when the column doesn't have duplicate values	Last value in rank column will be equal to no of records in the table when the last record in the table doesn't have duplicate value
It won't skip numbers	It will skip numbers

21) What is on delete cascade and on delete set null?

On delete cascade: when we try to delete data from parent table, system won't allow deleting if corresponding child record is found. If we want to delete data from parent table then we can use "on delete cascade". If we use 'ODC' in fk and if we delete data from parent table then automatically corresponding data from child table will be deleted.

On delete set null: when we try to delete data from parent table, system won't allow deleting if corresponding child record is found. If we want to delete data from parent table then we can use "on delete set null". If we use 'ODSN' in fk and if we delete data from parent table then automatically corresponding data from child table will be set as null.

22) Diff between procedure and trigger.

TRIGGER	PROCEDURE
Trigger fires implicitly	Procedure fires explicitly
We cannot give parameters to triggers	We can give parameters to procedure
Frontend resource cannot call triggers	Anyone can call procedure and use it
We cannot debug trigger	We can debug procedure
We can call procedure inside triggers	We cannot call triggers inside procedures
We can disable or enable triggers	We can drop procedure
Triggers are mainly used to avoid transaction, to maintain audit, to do one action while another action is going on	Procedure is mainly used to do calculations

23) Diff between procedure and package.

PACKAGE	PROCEDURE
Package is collection of related objects	Procedure is standalone object with set of statements
Package has 2 parts SPEC	Procedure has no parts. Complete procedure we can write in one area

BODY	
If we call any program in the package then entire package will come and store in the ram until the session end	If we call procedure, then the procedure will come and store in ram until program execution only
If we declare any variables in package spec then those variables will become global and we can use those global variables inside the package and outside the package	Procedure variables are local variables
If we want to declare any objects that can be done only in package spec, if we want to write any function or procedure, we can do it in package body.	In procedure both variable declaration and body will be in one area
Packages support function overloading, forward declaration, on time procedure	Those we cannot see in procedures
Packages are mainly used to develop a module	Procedures are mainly used to do calculations

24) Diff between implicit cursor and explicit cursor.

IMPLICIT CURSOR	EXPLICIT CURSOR
If we want to check the status of dml operations in plsql block, then we will use “implicit cursor”	Explicit cursor is used to fetch records in plsql block
Implicit cursor is used to know the status of ‘into clause’	Into clause is there to fetch record, but it will fetch only one record. If we want to fetch more than one record then we will use explicit cursor.
Implicit cursor has a key word called ‘SQL’	In explicit cursor whatever name we want, we can give
Implicit cursor has 3 attributes FOUND NOTFOUND ROWCOUNT	Explicit cursor has 4 attributes FOUND NOTFOUND ROWCOUNT ISOPEN
Implicit cursor has no types	Explicit cursor has 3 types BASIC CURSOR CURSOR FOR LOOP CUROSR FOR LOOP WITH SELECT STATEMENT

25) Diff between explicit cursor and ref cursor.

EXPLICIT CURSOR	REF CURSOR
If want to retrieve data of 10 different tables, then we need to write 10 diff explicit cursors.	If we want to retrieve data of 10 different tables, One weak ref cursor is enough to get the 10 different tables data.
Types of explicit cursor BASIC CURSOR CURSOR FOR LOOP CURSOR FOR LOOP WITH SELECT STATEMENT	TYPES OF REF CURSOR WEAK REF CURSOR STRONG REF CURSOR
We cannot pass explicit cursor as parameter or in return	We can pass ref cursor as parameter and in return also we can use ref cursor

Explicit cursor has no global type	Ref cursor has global type called sys refcursor
Explicit cursor always associates with same result set	Ref cursor can be assigned to different result sets
We cannot pass table name and column name dynamically	We can pass table name and column name dynamically
We need to declare initially query in declaration	No need declare query initially, rather we can pass query dynamically in runtime

26) Advantage of packages.

Advantages of packages:

- Package is used to develop module of an application and help us to maintain our application in more efficient way. If we call any packaged program then entire package will come and store in the ram until the session end, if we call any other program of the same package it will be executed immediately as the entire package has stored in the ram and no need to go and search in the database.
- If we declare any function/procedure/variable/collection/cursor in the package spec then these objects will be global. They can be used inside the package and outside the package.
- Package is collection of related objects.
- Package supports function overloading means we can write any no. of functions of procedures inside package with same name but the no. of parameters must be different.
- When we declare variables in package spec then they will become global variables.
- If we can drop package, then both spec and body will be dropped at a time. We cannot drop separately.

Disadvantages of packages:

- Without package spec we cannot write package body.
- It occupies space.
- Package won't support backward navigation.

27) Diff between raise and raise application error.

RAISE	RAISE APPLICATION ERROR
Raise needs 3 sections DECLARATION SECTION HANDLING SECTION EXCEPTION SECTION	RAE needs only one section HANDLING SECTION
Raise doesn't need any error number	RAE needs error number between -20000 to -20999
Raise will terminate the program there itself and enters the exception section	RAE will terminate the program there itself and shows error message on the screen.
Raise won't rollback the previous DML operations	RAE will rollback all the previous DML operations

28) Diff between predefined and non predefined exception.

PREDEFINED EXCEPTION	NON PREDEFINED EXCEPTION

When we get an error that was in the predefined errors, then we will use predefined exceptions	Oracle raise lakh's of errors except predefined errors, if we want handle all these errors then we will use non predefined exceptions
Predefined exceptions do not require declaration section	Non predefined exceptions need declaration section
We can handle predefined exceptions using predefined names	We will use pragma exception init with error number in Non predefined exceptions.

29) Diff between cursor and collection.

CURSOR	COLLECTION
Cursor fetches data record by record	Collection fetches all data at time
Cursor requires more no. of context switches to fetch data	Collection requires only one context switch to fetch data
Cursor has global type called sys refcursor	Collection has no global data type
In cursors, we can use normal variables and collection variables	In collection, we must use collection variable
Cursor has 3 types IMPLICIT CURSOR EXPLICIT CURSOR REF CURSOR	Collection has only one type BULK COLLECT
Cursor has attributes	Collection has methods
We can use into clause and bulk collect into clause	We can use only bulk collect into clause
Cursor supports only forward navigation	Collection supports both forward and backward navigation
There are 2 types of errors in cursors INVALID CURSOR CURSOR IS ALREADY OPEN	There are 4 types of errors in collection NO DATA FOUND SUBSCRIPT BEYOND COUNT SUBSCRIPT OUT OF LIMIT REFERENCE TO UNINITIALIZED COLLECTION

30) What is bulk bind?

Bulk Bind: oracle uses two engines to execute plsql code. All the program code is handled by plsql engine and all the sql statements are handled by sql engine.

When we write any dml statement in collection loop then it will be considered as separate statement. So to execute each dml statement separate context switch is required, to avoid this we will use bulk bind.

Bulk bind will wait till the end of loop, bind all the dml statements and in one shot it will send them to sql engine.

31) What is bulk limit?

Bulk Limit: plsql collections are in memory. Massive collections may affect the performance of the system due to huge amount of data. So to avoid this we will use bulk limit.

Bulk limit will chunk the huge data of collection and process each chunk in one turn.

Bulk limit we have to write along with collection only.

32) What is cursor 'for update'?

For update: if we write ‘for update’ in cursor select statement then exclusively row level lock will be applied for all the rows which are identified by cursor select statement.

Holding these records for our changing purpose only, until we give commit or rollback, no one else will be able to do any changes for these records.

33) What is where ‘current of’?

Where current of: when we plan on deleting or updating the records which are identified by cursor select for update statement, “where current of” statement allows us to update or delete the record which is last fetched by the cursor.

34) What is “mutating error”?

Mutating error: mutating error occurs when a row level trigger is trying to apply dml operations or select statements on the same table, while trigger is firing.

Ex: trigger is firing on the emp table on before update case and inside trigger body has select statement on the same emp table, then mutating error will occur.

35) What are the advantages and disadvantages of triggers?

Advantages:

- To avoid transaction
- To maintain audit
- When any transaction is going on and at the same time if we want another transaction to be done automatically then we will use trigger.

Disadvantages:

- We can’t debug trigger.
- We can’t give parameters to triggers.
- We can’t call triggers inside other programs like functions, procedures, packages etc.
- Triggers can’t be fired explicitly.
- Front end resources can’t call triggers
- When we drop the related table, then trigger will also dropped.
- We can’t write TCL commands inside triggers.

36) What are the disadvantages of packages?

Disadvantages:

- When call any packaged program then entire package will come and store in the ram until the session ends. Even if we don’t call any other program of the package still it will be exists in the ram until the session ends. So wastage of memory will be there.
- We cannot write package body without package spec.
- We need to write both package spec and package body. If we want to declare any variables we can do it in package spec and if we want to write any function/procedure then we will write it in package body.

37) What are the disadvantages of functions?

- If we write dml operations inside functions, then we can’t call those functions inside sql statements. So, if we want to call those functions inside sql statements then we need pragma autonomous transaction+ TCL command.

- If we use out parameter inside a function, then at any cost we cannot call that function inside sql statement. There is no medicine for this.
- When we call function in plsql block then we need a variable as function must return a value.

38) What are the disadvantages of procedures?

- We cannot call procedures inside sql statements.
- If we want output from procedure then we need to use out parameter.

39) Difference between external table, database table and global table.

EXTERNAL TABLE	DATABASE TABLE	GLOBAL TEMPORAY TABLE
It won't store data physically	It will store data physically	It will store data physically
It won't require table space	It requires table space	It has default table space called temporary table space
We can't create indexes, triggers, constraints on external table	We can create triggers, indexes, constraints on database tables	We can create triggers, indexes, constraints on GTT
If we export external table then we will get only table structure	If we export database table then we will get both structure and data of the table	If we export GTT then we will get only table structure
We can't apply DML operations on external tables	We can apply DML operations on database tables	We can apply DML operations on GTT
If we want to modify data in external table then we need to modify the flat file	If we want to modify data in database table then we need to apply DML operations	If we want to modify data in GTT table then we need to apply DML operations

40) What is bulk collect?

BULK COLLECT: in order to reduce no. of context switches between SQL engine and PLSQL engine, we will use 'bulk collect'.

Bulk collect will collect all the data from SQL engine and in one shot it will send all the data into collection variable using only one context switch. (Collection variable means composite data type variable) or

SQL engine gets data from database and in one shot bulk collect assign the whole data to the collection variable using only one context switch.

Always we need to write bulk collect along with select statement.

41) What is bulk exception? Write an example.

Bulk exception: when we use bulk bind we need to multiple operations in single shots, then there is chance to get bulk of errors at one time. If we want to catch all those errors then we will use 'bulk exception'.

Its error number is -24381.

Bulk exception can be used along with bulk bind concept only.

42) Why dynamic SQL is required? Where did you use in your project?

Dynamic SQL: oracle won't allow us to use DDL commands inside PLQSL block, if we want to use DDL commands inside PLSQL block then we needs to use "EXECUTE IMMEDIATE" command.

Execute immediate is dynamic SQL. We will use dynamic SQL in two cases.

Case 1): to use DDL commands inside PLSQL block.

Case 2): to pass table names or columns name dynamically at runtime, for DML operations also.

43) What is pragma serially reusable?

Pragma serially reusable: generally when we call any packaged program then entire package will come and store in the ram until the session end. If we write "pragma serially_reusable" inside the package spec, then the package will become one time call package. Means it won't be stored in the ram and immediately memory will be cleared from the ram.

44) What is pragma autonomous transaction?

Pragma autonomous transaction: it defines the independent transaction from the parent transaction and allows commit or roll back without affecting the parent transaction.

We can use PAT in

- Error log table
- Commit inside trigger
- To call function with DML operations inside select statement

45) Diff between predefined and user defined exception.

Predefined exception	User defined exception
Pre defined exceptions don't require any declaration section	User defined exceptions like raise requires declaration section
There are 14 predefined errors like no data found, too many rows, zero divide, and invalid cursor etc..,	User defined exceptions are only 2 types 1) RAISE 2) RAE
We can write predefined exceptions anywhere we want but 'others' exceptions must be at the end of all exceptions	We can write user defined exceptions anywhere we want in the program
Oracle has provided freely some errors those are called oracle genuine errors, we can use those. When we get oracle genuine errors and if we want to handle those errors then we will use predefined exceptions	It is not oracle genuine error, based on user or business requirement if we want to raise error then we will use user defined exceptions Ex: icici bank should have min bal of 10000, if bal<10000 then raise error

46) Why exception is required?

An error occurred during execution of program is run time error in PLQSL. PLQSL facilitates programmers to catch such run time error conditions using exception block in the program and an appropriate action is taken against the error condition.

47) What is function overloading?

Function overloading: in a package if we want two or more functions or procedures with same name we can write but the no. of parameters must be different. It is called function overloading.

48) What is forward declaration?

Forward declaration: if we declared any private program inside package body and we called that private program in another program in the same package body, then that private program must be declared before the calling program. It is called forward declaration; it will be applicable only for private programs

49) What is one time procedure?

One time procedure: if we write begin and set of statements at the end of package body, then it s called 'one time procedure'. It has no end statement.

If we test any packaged program then it will first enter into package spec part and then it enters into one time procedure after that it will test the packaged program.

One time procedure is like gateway to package. One time procedure must be declared at the end of the package only.

50) Explain about global temporary table.

Global temporary table: GTT store data physically. GTT has table space by default temporary table space. GTT is of 2 types:

Statement level GTT: if we use 'on commit delete rows' at the end of GTT creating statement and when we insert data into GTT and then give commit, the data in GTT will be deleted. This in known as 'STATEMENT LEVEL GTT'.

Session level GTT: if we write 'on commit preserve rows' and when we insert data and commit, it will store data for that session only. This is known as 'session level GTT'.

When we export data in GTT then we will get only table structure. If we want to change the data in the GTT then we need to apply DML operations.

51) Explain about parameter methods?

There are three types of parameters

1) In : through the program we cannot change the value.

In parameter is mainly used for the end users (customers).

2) Out : when we need to display output on the screen after calculations we will use out parameter (means when user expecting some value on the screen then we will use out parameter).

Whatever value we want we can give throughout the program.

3) In out: for which parameter user is passing value same parameter will give the output.

Throughout the program we can give any value.

Mainly used for collection types.

52) Explain about cursor attributes?

Implicit cursor: implicit cursor has 3 attributes.

1) Found: it is a Boolean value, returns true or false.

If no select or DML statement is run, it will return null.

If the most recent select or DML statement fetches record then it will return true.

If the most recent select or DML statement fetches no record then it will return false.

2) Notfound: it is a Boolean value, returns true or false.

If no select or DML statement is run, it will return null.

If the most recent select or DML statement fetches record then it will return false.

If the most recent select or DML statement fetches no record then it will return true.

3) Rowcount: it returns number.

If no select or DML statement is run, it will return null.

If the most recent select or DML statement fetches record then it will return number of rows fetched so far.

Explicit cursor: explicit cursor has 4 attributes.

1) Found: it is a Boolean value, returns true or false.

If no select or DML statement is run, then cursor_name%found will return null.

If the cursor last fetch returned a row then cursor_name%found will return true.

If the cursor last fetch won't returned a row then cursor_name%found will return false.

2) Not found: Found: it is a Boolean value, returns true or false.

If no select or DML statement is run, then cursor_name%found will return null.

If the cursor last fetch won't returned a row then cursor_name%found will return true.

If the cursor last fetch returned a row then cursor_name%found will return false.

3) Row count: it returns number.

If the cursor last fetch returned a row then cursor_name%found will return number of rows, it will be incremental.

4) Is open: if will return true if the cursor is opened else it will return false.

53) Explain about collection methods

54) What are the errors in triggers? When those errors will be occurred?

Errors in triggers: generally we will get mutating error and recursive error in triggers.

Some times we will get dead lock error also (when we use pragma autonomous transaction + tcl in triggers).

Mutating error: when a row level trigger tries to do any DML operation or select statement on the same table while trigger is firing on that same table.

To resolve mutating error we will use pragma autonomous transaction +TCL command.

If we use PAT + TCL then there is chance to get dead lock error.

Recursive error: when we write a DML action in the trigger and we are using same action inside trigger body then we will get recursive error.

Note: recursive error in row level trigger by insert and insert

Statement level trigger by insert insert update update delete delete.

Ex: trigger is firing on emp table before insert case, inside trigger body insert statement is used on the same table. Then we will get recursive error.

55) What are the errors in collections? When those errors will be occurred?

Errors in collections:

1) No data found: mostly we will see no_data_found error in PLSQL table. Sometimes we will this error in nested and varray table.

Ex: PLSQL table has 5 records and if we try to fetch 6th record from that collection, then we will get NO_DATA_FOUND error.

If nested or varray table has 5 records and we have deleted the 2nd record and again if we try to fetch 2nd record then we will get NO_DATA_FOUND error.

2) Subscript beyond count: we never get this type of error in PLQSL table.

If nested or varray has 5 records and if we try to fetch 6th record then we will get subscript beyond count error.

3) Subscript out of limit: we will get this type error only in varray table.
   Ex: if the VARRAY has size of 10 and if we try to store 11th record into the VARRAY then we will get subscript out of limit error.

4) Reference to uninitialized collection: we don't get the error in PLSQL table.
   When we try to use methods inside uninitialized collection then we will get reference to uninitialized collection error.

56) What are the errors in cursors?

Errors in cursors:

1) Invalid cursor: when user tries to fetch data without opening cursor or when user tries to open cursor with wrong name then we will get invalid cursor error.
2) Cursor is already open: when we try to open the cursor which is not closed then we will get cursor is already open error.

57) We have 'when others' exception? Then why we are using other predefined exceptions?

When we use 'when others' exception instead of all other predefined exceptions, then we will not be able to take appropriate action against that particular error condition.

58) What is materialized view?

Materialized view: materialized view is database object which consists of results of the query. Materialized view stores the select statement and the data. It will not impact on the original object while performing DML operations. If we want those DML operations to impact on the original object then we need to use some methods in materialized view.

59) Difference between normal view and materialized view.

Normal view	Materialized view
View is stored select statement	Materialized view is database object which contains result of a query.
View is logical component	Materialized view is physical component
When any DML operation is performed on the view, it will impact the original object and vice versa.	When any DML operation is performed on MV that won't impact the original object. But when any DML operation is performed on the table it may or may not impact the MV and depends on the type of MV.
View won't store data physically	MV stores data physically
We can't create triggers, constraints, indexes on views	We can create triggers, constraints, indexes on MV
View doesn't require any table space	MV need table space
When we drop any column from the table on which basis view is created then view will become invalid	When we drop any column from the table on which basis MV is created then it won't impact MV

60) What is use of returning clause?

Returning clause: returning clause is used to give the status or return of the DML value, no need to write select statement. At the time of DML operation, there itself we can return the effected values by using returning clause.

The returning value may be collection type or single value. Both we can return in this returning clause.

61) What is cascade constraint?

62) What is %type and %rowtype?

% type: %type is nothing but column type, we can declare the variable according to column structure. If future if there is any increase or decrease in size of column or change in data type but dynamically this variable can be exists.

Syntax: variable_name table_name.column_name%type;
Ex: lv_ename emp.ename%type;
%rowtype: %rowtype means we can declare the variable according to row structure. In future when any column is added or dropped but that variable can take whatever changes happened.
Syntax: variable_name table_name%rowtype
Ex: ved emp%rowtype;

63) What is explicit cursor?

By using select into clause we can fetch only one record in PLSQL block, if we want fetch more than one record in PLSQL block then we will use explicit cursor. When we execute any PLSQL code that will be executed by PLSQL engine, when any SQL statements are found in PLSQL BLOCK, immediately they will be sent to SQL engine and those SQL statements are executed by SQL engine. To get data fetched by SQL engine to PLSQL engine by using context switches, PLSQL creates an area to store the data. That is called cursor area. (or)

Cursor is a private area created by PLSQL engine, to store the data fetched by SQL engine in the PLSQL block.

Explicit cursor has 4 attributes:

1) %found
2) %notfound
3) %rowcount
4) %isopen

64) What is implicit cursor?

Implicit cursor will give the status of DML operation and select query in PLSQL block. Implicit cursor will be automatically created by PLSQL engine when user executes DML or select in PLSQL block. Implicit cursor has predefined name called 'SQL'. Implicit cursor has 3 attributes

1) %found
2) %notfound
3) %rowcount

65) What is trigger? And when we use triggers?

Trigger is explicit object, but it will be fired implicitly when event matches or based on event.
Triggers are mainly used

1) to avoid/ to restrict transaction
2) to maintain audit or history
3) When one action is going then if we want to perform another action automatically.

66) Difference between before and after in triggers.

BEFORE	AFTER
If there is before RLT and we have executed DML then first Before RLT is fired and then DML statement executes.	If there is after RLT and we have executed DML then first DML statement executes and then after RLT is fired.
We can catch or change old and new values in BRLT	We can catch old and new values in ARLT, but if we try to change old and new in ARLT then program will be compiled with errors

Data won't be stored in buffer (means it won't occupy space in memory why because before DML action trigger will be fired)	Data will be stored in the buffer (means it will occupy space in the memory)
If table has unique constraint and BRLT, if we try to insert duplicate values then first BRLT will be fired	If table has unique constraint and ARLT, if we try to insert duplicate values then first constraint will be fired.
BRLT is best for RAE	ARLT is best for audit maintenance

67) Difference between statement level and row level triggers.

Statement level trigger	Row level trigger
SLT fires only once for the entire table(irrespective of no. of rows effected)	RLT fires based on no. of rows effected(if n no. of rows effected then n times RLT will fire)
If table has '0' records still STL will fire once	If table has '0' records RLT won't fire
We can't catch old and new values	We can catch old and new values
In SLT we will get only one error i.e recursive error	In RLT we will get recursive error and mutating error. And in rare cases we will get dead lock error
SLT doesn't have any other extra clauses	RLT has one extra clause 'for each row'
Main purpose of SLT is to avoid / restrict transaction	Main purpose is to maintain audit

68) What is the use of package spec without package body?

When we declare any variables/functions/procedures/cursor etc in package spec then they will become global means we can use them inside package body or outside anywhere.

69) What is merge statement?

By using merge statement we can perform more than one DML operation at one time on a table (based on another table).

Syntax: merge into target_table using source_table

On condition

When matched then condition

When not matched then condition

70) Difference between DDL and DML triggers?

DDL	DML
DDL triggers are schema level	DML triggers are object level triggers
DDL triggers are only one type	DML triggers are 3 types 1) Statement level triggers 2) Row level triggers 3) Instead of triggers
In DDL triggers we should use pseudo columns to raise error on particular object, for particular type of action	DML trigger always used to write only on one object, on which operation we want to put condition that we can write directly in trigger
In DDL triggers we won't get any type of errors	In DML triggers we will get errors like Mutating error Recursive error Sometimes dead lock error

In DDL triggers we will use pseudo columns only	In DML triggers sometimes we will use old and new values

71) Explain about each exception.

Predefined exception when we get error in oracle genuine errors then we will use predefined exception. Oracle freely provided 14 predefined errors. By using those predefined error names we will handle errors. But, in predefined errors, ‘when others’ must be at the end of all the exceptions.

User defined exception when we want to raise any error if a particular condition matches or based on business requirements when we raise any error. And if we want to do something when a particular condition matches then we will use ‘user defined exceptions’. User defined exceptions are 2 types

1) Raise
2) Raise application error

Raise needs 3 sections 1) declaration section 2) handling section 3) exception section

Raise will terminate the program there itself and enter the exception section.

Rae needs only one section that is handling section. Rae will terminate the program there itself and throw and error message on the screen. And it will rollback all the previous DML operations.

Rae needs error number between -20000 and -20999.

Non predefined exception/ user named exception oracle will raise lakh’s of errors which are not in predefined errors. If we want to handle all those errors then we will use non predefined exception. Non predefined exceptions requires to sections 1) declaration section 2) exception section. By using pragma exception_init along with exception name and error number in the declaration section we will use non predefined exceptions.

72) Explain about joins.

To fetch data from more than 2 tables then we use joins. We can join any no. of tables but if we join n tables then at-least n-1 conditions are required.

Nulls will never join.

Joins are mainly used to avoid Cartesian product.

Types of joins

1) equi-join
2) Non equi join
3) Outer join a) left outer join b) right outer join c) full outer join
4) Cartesian join
5) Self join (explain about each join with one example for each)

73) What are index scans?

Index scans are used to avoid table access full and collect data from table with in less time than table access full.

Table access full is nothing but system blindly go and searches all the rows without knowing the address or if the index is not selective then also system will go to table access full.

System will decide to use index or not based on selectivity.

Selectivity = total distinct values / total values

Selectivity	Index
0-0.5	Bitmap
0.5-1	B-tree
1	b-tree / unique

If index is selective then only system will go to index scans otherwise it will skip index and go to table access full.

Types of index scans

1) Unique index scan

Select * from emp where empno=7369—if empno has unique index

2) Index range scan

Select * from emp where empno=7369—if empno has b-tree index

3) Index fast full scan

Select empno from emp

4) Index full scan

Select empno from emp order by sal

5) Index skip scan

Here if the index is not selective index then system will skip the index and goes to table access full.

74) What are join methods?

Oracle internally performs join methods when user executes any join query. If user executes any joining query then oracle internally fetches data by using these join methods.

1) Nested loop join
2) Sort merge join
3) Hash join (explain about each join methods which relative examples)

75) What is TK PROOF?

To check the performance of the application, user generates trace files. But trace files are in system readable format, to convert system readable trace files into user readable TK proof is used.

Trace Files: oracle will save the information in trace files for last 30 mins or for last 2 hours, how many programs are executed and how much time they took to execute.

Alter system set SQL_TRACE=true; ---in command window

True – start trace files

False – stop trace files

76) What is cost, cardinality, bytes?

Cost – no. of hits to database (means how many times system goes to database to fetch data)

Cardinality – no. of rows searched in database to fetch the data

Bytes – how much area searched in database to fetch the data.

77) What is index organized table?

When we confidently know that based on only one column we will search/filter data by using that column in where condition, then we will use index organized table. But that column should contain primary key. Index organized table stores the entire table data in the tree structure along with their rowids. But when user searches data by using any other column in where condition then entire index organized table is completely failure.

Ex: Create table emp(employee no number, name varchar2(50), salary number constraint pk_empno primary key(employee no) organization index)

78) What is partition table?

Partition divides large table data into more small manageable pieces. Partition allows tables, indexes and index organized tables to be sub divided into small pieces.

If table having more than 2gb of data then only we will use partitions. When contents of table need to be distributed across different storage devices.

Types of partitions 1) range partition 2) list partition 3) hash partition

(Explain about each partition in detail with examples)

79) What is re-map data masking?

remap_data parameter will allow to specify a remap function that takes as a source the original value of the designated column and returns a remapped value that will replace the original value in dump file. A column use for this option is to Mask data when moving from a production system to a testing system.

80) What is hint?

Hint will pass a message on how a query should execute.

/*+ idx_empno(empno)*/

Normal hint: select empno /*+ idx_empno(empno)*/, ename from emp

Parallel hint: select empno /*+parallel (50)*/ , ename from emp

No copy hint: it will pass values directly from actual parameter to formal parameter without storing in the ram. Generally no copy hint will be used for out and inout parameters mostly for collection variables .

81) What is optimization?

Optimization will plan the execution of a query like which database file, which extend, which join method, which block and which index scan system use for execution of a query. Once user runs a query, system optimization automatically generates plan id. If the same query is executed again system brings the output within less time than previous execution by using plan id.

82) What is normalization?

Normalization is the process of combining various individual modules into single process by using data integrity, main purpose is to avoid data redundancy (means to avoid duplicate).

Data integrity is nothing but integrity constraint (foreign key).

83) What is index? Why index is required? What are types of indexes?

Index will point out the physical address of the data. While retrieving or manipulating data index will help to do it fast. Or index is a database object which is used to retrieve or manipulate data fast.

Index will store the rowids only not the entire table data. Index will be stored in one area along with indexed column data and rowids and entire table data will be stored in other area. Index will store data in the form of tree structure or table format based on the type of index.

Types of indexes

1) B-tree index/ binary index
2) Unique index
3) Bitmap index
4) Function based index

5) Reverse key index(explain about each index with examples and also explain about rebuild of index and online key in index).